

eventual personal usage of energy is mostly done in the DC form. Hence, looking at various aspects, decentralised energy storage systems become very important.

Portable power stations (PPS) are a new breed of products that allow significant energy storage at the user end and also allow for its use in either AC or DC form. The PPS come in various shapes and sizes, from the small ones giving about 50Wh of energy to larger stations of up to 4kWh capacity, which provide on-demand energy at customer establishments to run various appliances for a significant time.

Being an energy storage equipment, it obviously comes with the relevant battery storage and battery management systems. The PPS, however, also come with special capabilities which differentiate them from simpler equipment, such as portable power banks. These have following abilities:

1. Provide both AC and DC forms of power and sometimes at multiple voltages.
2. Can get charged from multiple sources (in some cases simultaneously), such as the grid AC supply or via direct DC inputs like solar PV or in-vehicle 12V supplies.
3. Can be paralleled to provide for on-demand larger storage capabilities.

In addition to their ability to provide user-end on-demand energy, these systems can also play a pivotal role in times of natural disasters. As we all know, the first things to go out when a storm hits your neighbourhood is the electrical supply. In such times, a PPS can be a life saver, as it can run most of your home equipment for a long time. PPS units can also be lifesavers on long trips, off roading adventures, and other places, such as family picnics and camping.

As electrical vehicles become more common, PPS units can be

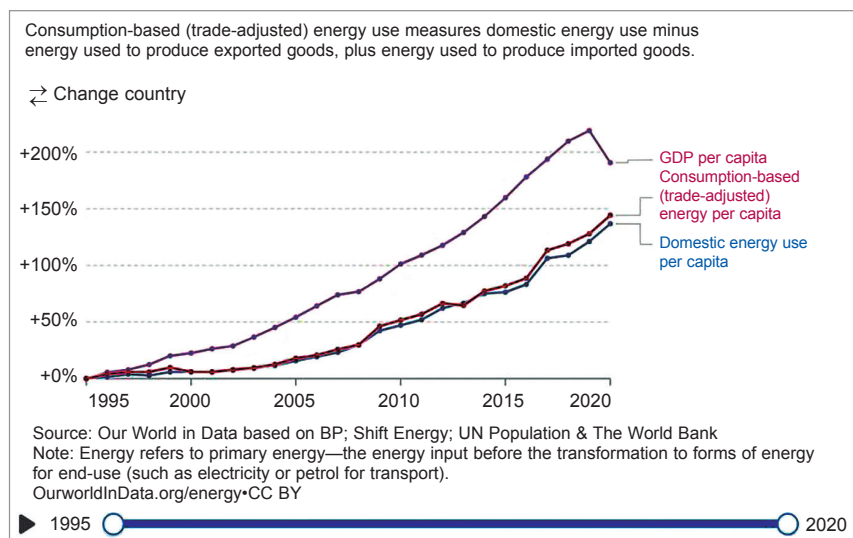


Fig. 2: Change in energy use vs GDP per capita for India (Source: <https://ourworldindata.org>)

treated as your reserve fuel, as they can charge the EV through their AC outlets. Those from an earlier generation may remember portable generators, which could be used to provide on-demand power in shops, markets, and homes etc. All such use cases apply directly for portable power stations.

For the benefit of all, let's try to figure out what goes into making a portable power station. As the name suggests, a PPS is a store house of energy that can provide significant amounts of power through its various different output ports and can be charged up from various sources of electrical energy.

Some of the well-known brands of PPS in the USA are:

1. Jackery
2. EcoFlow
3. Bluetti
4. Anker
5. Pectron

Solar products suppliers in India also provide variations of PPS that are sometimes called solar generators, or solar home UPS, or solar backup home lighting systems. Most of these are stationary systems installed in homes and offices and used mostly for lighting purposes. The aspect of portability while pro-

viding a significant power output is rarely available and presents a Blue Ocean market situation for Indian manufacturers.

Besides, while there are multiple solar home lighting system suppliers in India, their products often look and feel similar, which suggests that these products are assembled from commonly available parts and not manufactured by the supplier. The actual look and feel of a product play a big role in forming a connect with the customer and this aspect should be considered by the manufacturers.

In addition, there are many white good brands that also produce PPS. The above US brands have been able to make a name for themselves by providing a number of choices to the customers, so that products that suit their exact needs are made available to them. Examples of these products are shown in Fig. 3.

As shown in Fig. 4, a PPS consists of mainly the following:

1. AC-DC converters
2. DC-DC converters connected to DC inputs
3. Multiple input charge controller
4. Battery source with a BMS